

Paramount Diagnostic Lab

NT-proBNP

Test Details

METHODOLOGY

Electrochemiluminescence immunoassay (ECLIA)

RESULT TURNAROUND TIME

1 - 2 days

Turnaround time is defined as the usual number of days from the date of pickup of a specimen for testing to when the result is released to the ordering provider. In some cases, additional time should be allowed for additional confirmatory or additional reflex tests. Testing schedules may vary.

Use

This assay is used as an aid in the diagnosis of individuals suspected of having congestive heart failure. The test is further indicated for the risk stratification of patients with acute coronary syndrome and congestive heart failure. The test may also serve as an aid in the assessment of increased risk of cardiovascular events and mortality in patients at risk for heart failure who have stable coronary artery disease.

Special Instructions

This test may exhibit interference when sample is collected from a person who is consuming a supplement with a high dose of biotin (also termed as vitamin B7 or B8, vitamin H, or coenzyme R). It is recommended to ask all patients who may be indicated for this test about biotin supplementation. Patients should be cautioned to stop biotin consumption at least 72 hours prior to the collection of a sample.

Limitations

NT-proBNP values should be assessed in conjunction with other cardiovascular risk factors and clinical findings.

Interpreting results in the setting of comorbidities such as cardiac, pulmonary, and renal disease, which are likely to increase natriuretic peptides above current thresholds for HF, should be done with caution.¹

The accuracy of natriuretic peptides for the detection of HF is reduced in the setting of atrial fibrillation and sepsis, and careful interpretation is warranted.¹

Custom Additional Information

Left ventricular dysfunction can occur as a part of coronary heart disease, arterial hypertension, valvular disease, and primary myocardial disease. If the left ventricular dysfunction remains untreated and is progressive, the potential for mortality is high ,e.g. due to sudden cardiac death. Chronic cardiac insufficiency is a clinical syndrome caused by impairment of the cardiac pumping function. Based on the symptoms, the severity of cardiac insufficiency is classified in stages (New York Heart Association classification [NYHA] I-IV).^{3,4} Clinical information and imaging procedures are used to diagnose left ventricular dysfunction.⁵

The significance of natriuretic peptides in the control of cardiovascular system function has been demonstrated. Studies reveal that natriuretic peptides can be used for diagnostic clinical problems associated with left ventricular dysfunction.⁶ The following natriuretic peptides have been described: atrial natriuretic peptide (ANP), B-type natriuretic peptide (BNP), and C-type natriuretic peptide (CNP).^{7,8} ANP and BNP, as antagonists of the renin-angiotensin-aldosterone system, influence by means of their natriuretic and diuretic properties, the electrolyte and fluid balance in an organism.^{9,10} In subjects with left ventricular dysfunction, serum and plasma concentrations of BNP increase, as do the concentrations of the biologically inactive prohormone, proBNP. ProBNP, comprising 108 amino acids, is secreted mainly by the ventricle and, in this process, is cleaved into physiologically active BNP (77-108) and the N-terminal fragment NT-proBNP (1-76).⁸

Studies indicate that NT-proBNP can be used in diagnostic and prognostic applications.¹¹⁻¹³ The concentration of NT-proBNP in serum or plasma correlates with the prognosis of the left ventricular dysfunction. Fisher et al found that congestive heart failure patients with NT-proBNP values above median had a 1-year mortality rate of 53% compared to 11% in patients below median.¹⁴ In the GUSTO IV study, which involved more than 6800 patients, it was shown that NT-proBNP was the strongest independent predictor of 1-year mortality in patients with acute coronary syndrome.¹⁵

The following cut-points have been suggested for the use of proBNP for the diagnostic evaluation of heart failure (HF) in patients with acute dyspnea^{16,17}:

Modality	Age (years)	Optimal Cut Point
Diagnosis (rule in HF)	<50	450 pg/mL
	50 - 75	900 pg/mL
	>75	1800 pg/mL
Exclusion (rule out HF)	Age independent	300 pg/mL

Specimen Requirements

Specimen

Serum

Volume

1 mL

Minimum Volume

0.7 mL (**Note:** This volume does **not** allow for repeat testing.)

Container

Gel-barrier tube (preferred) or red-top tube

Collection Instructions

If a red-top tube is used, transfer separated serum to a plastic transport tube.

Stability Requirements

Temperature	Period
Room temperature	3 days
Refrigerated	6 days
Frozen	24 months
Freeze/thaw cycles	Stable x1

Reference Range

See table.²

Age	Male (pg/mL)	Female (pg/mL)
0 to 11 m	Not established	Not established
1 to 3 y	0–320	0–320
4 to 6 y	0–190	0–190
7 to 9 y	0–145	0–145
10 y	0–112	0–112
11 y	0–317	0–317
12 y	0–186	0–186
13 y	0–370	0–370
14 y	0–363	0–363
15 y	0–217	0–217
16 y	0–206	0–206
17 y	0–135	0–135
18 y	0–115	0–115
19 to 44 y	0–86	0–130
45 to 54 y	0–121	0–249
55 to 64 y	0–210	0–287
65 to 74 y	0–376	0–301

Age	Male (pg/mL)	Female (pg/mL)
>74 y	0–486	0–738

Storage Instructions

Room temperature

Causes for Rejection

Citrate plasma specimen; grossly hemolyzed samples; improper labeling

References

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Footnotes

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